From the properties of Miller’s planet, as depicted in *Interstellar*, a physicist who knows Einstein’s relativistic laws can deduce Gargantua’s mass and spin, and thence all else about it. Let's see how this works.[14]

Gargantua’s Mass

Miller’s planet (which I talk about at length in Chapter 17) is about as close to Gargantua as it can possibly be and still survive. We know this because the crew’s extreme loss of time can only occur very near Gargantua.

At so close a distance, Gargantua’s tidal gravitational forces (Chapter 4) are especially strong. They stretch Miller’s planet toward and away from Gargantua and squeeze the planet’s sides (Figure 6.1).